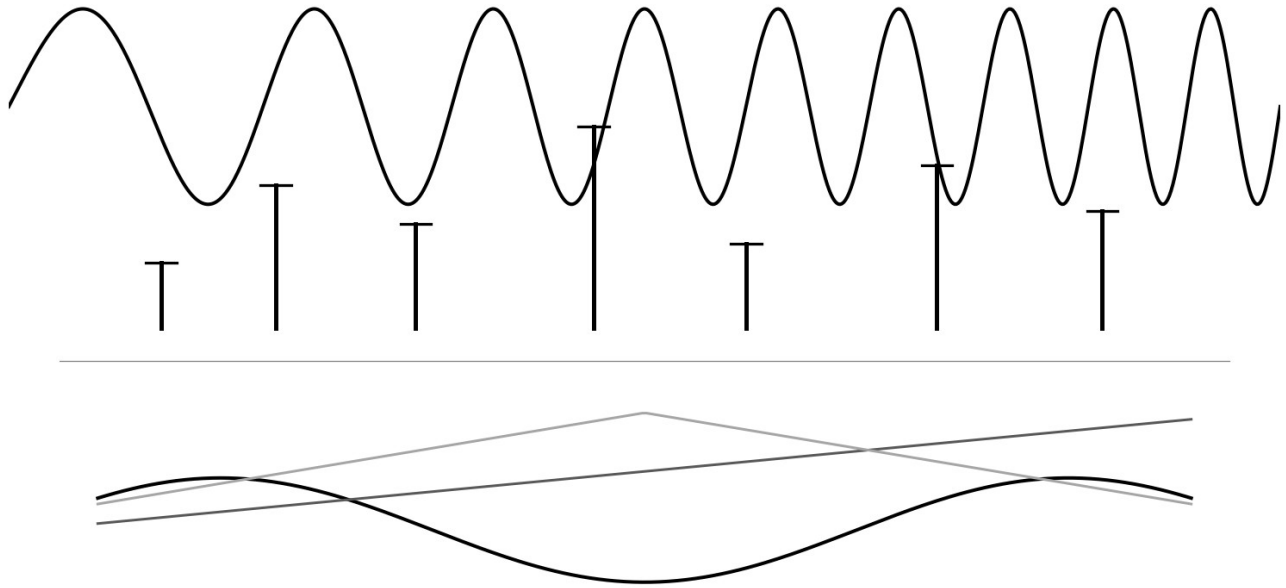


SPECTRAL ESTIMATION AND TIME-FREQUENCY ANALYSIS

Non-parametric analysis of dynamic and non-stationary signals



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Package structure. The ZIP archive contains the four MATLAB source files, this report, the generated figures and a compressed MAT file with independently reproduced reference results.

1. Abstract and objectives

1.1 Abstract

This practice develops three classical non-parametric tools for spectral analysis without relying on MATLAB built-in estimators: a window-normalized periodogram, the Welch averaged periodogram and a short-time power representation. The algorithms are applied to a complex synthetic record that combines stationary, intermittent and frequency-modulated components. The experiment demonstrates why a single global spectrum is insufficient for non-stationary signals and how the short-time Fourier transform reveals their evolution in time.

1.2 Objectives

- Implement the periodogram from segmentation, windowing, FFT magnitude and window-energy normalization.
- Implement Welch averaging with 50 percent overlap and quantify its variance-reduction effect.
- Build a short-time spectral matrix using a selectable time hop and frequency spacing.
- Generate a controlled non-stationary test signal whose theoretical frequency tracks are known in advance.
- Interpret the differences between time-averaged and time-resolved spectral representations.

1.3 Deliverables

The executable files use the MATLAB .m extension. The archive also includes `reference_results.mat` because .mat is MATLAB's binary data format rather than its source-code format. Running the main script produces an additional `practice_results.mat` file in the current MATLAB folder.

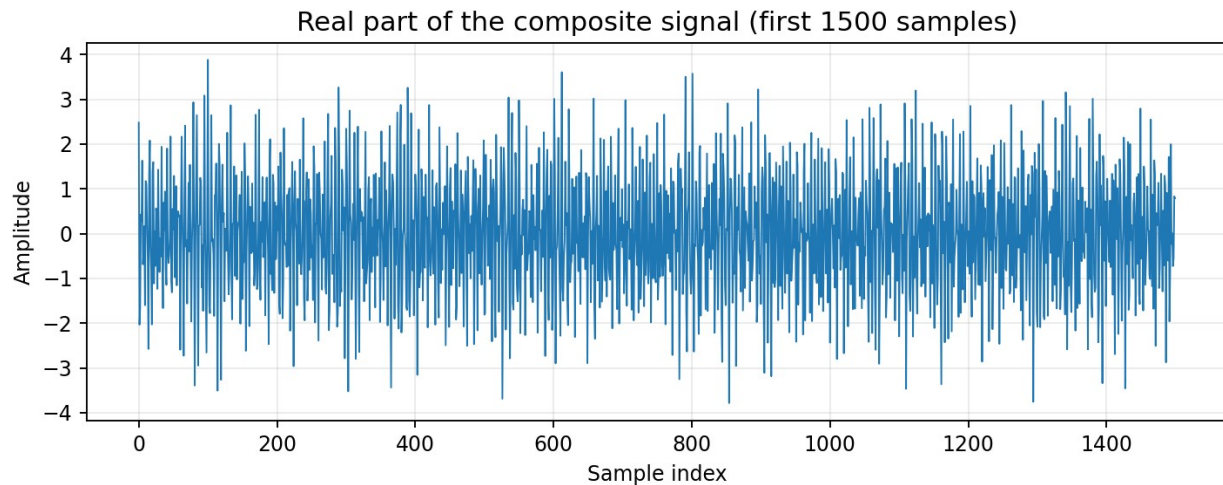


Figure 1. Short excerpt of the complex composite signal, represented through its real part.

2. Theoretical foundations

2.1 Windowed periodogram

For a length- N segment $x[n]$ and an analysis window $w[n]$, the implemented two-sided estimate is

Definition. $P_x[k] = |\text{FFT}\{x[n]w[n]\}|^2 / U$, where $U = \sum |w[n]|^2$. The normalization compensates for the energy introduced by the window. Since no sampling-frequency factor is included, the output is a power spectrum on the discrete FFT grid, not a density expressed per hertz.

The periodogram offers the full N -bin resolution of the selected segment, but it has high statistical variance. Its result also depends on which portion of a non-stationary record is selected.

2.2 Welch averaging

Welch's method divides the complete record into overlapping segments, applies the same window to every segment, evaluates one periodogram per segment and averages the results. Averaging reduces random fluctuations and produces a more stable global spectral description. The cost is the loss of temporal localization and some correlation between overlapping segments.

2.3 Short-time analysis

The short-time representation repeats the windowed periodogram at successive time positions. Each matrix column describes one time frame, while each row corresponds to one FFT bin. The analysis parameters obey $df = 2\pi/N$ and $\text{overlap} = N - dt$. Increasing N improves frequency resolution but makes rapid temporal changes harder to separate; decreasing dt increases time sampling and computational cost.

2.4 Hann window

The symmetric Hann window smoothly forces the segment edges towards zero. This strongly reduces spectral leakage compared with a rectangular truncation, at the expense of a wider main lobe. The source code constructs the window explicitly as $0.5 - 0.5 \cos(2\pi n/(N-1))$.

3. Custom periodogram

3.1 Processing sequence

1. Convert the input signal and window to column vectors.
2. Select the first N samples.
3. Multiply the segment by the N-sample analysis window.
4. Compute the N-point FFT.
5. Square the FFT magnitude.
6. Divide by the energy U of the window.

3.2 Core implementation

```
x_segment = x(1:N);
x_windowed = x_segment .* w;
X_windowed = fft(x_windowed, N);
U = sum(abs(w).^2);
Periodogram = abs(X_windowed).^2 / U;
```

3.3 Input protection

The final function checks the number of arguments, vector shape, positive integer FFT length, sufficient signal length, window length and non-zero window energy. These checks convert silent dimension mistakes into explicit diagnostic messages.

3.4 Interpretation

The function returns all N FFT bins because the experiment uses complex exponentials and therefore requires a two-sided digital-frequency representation over 0 to 2π . No one-sided doubling is performed. The main script displays the x-axis in units of π rad/sample, so FFT bin k is located at $2k/N$ on that axis.

Expected fixed-tone bins. For $N = 1024$, S1 appears at $k = 1024(50/128) = 400$ and S2 at $k = 1024(105/128) = 840$. The latter is equivalent to negative bin -184 in a centered spectrum.

4. Welch estimator

4.1 Segmentation

The selected segment length is $N = 1024$ and the overlap is $O = 512$, producing a hop of 512 samples. With a record of 32768 samples, the implementation uses 63 complete segments. Incomplete trailing segments are discarded, which is the usual Welch convention; a record shorter than N is zero-padded once.

4.2 Averaging algorithm

```
hop = N - O;
start_indices = 1:hop:(length(x) - N + 1);
for k = 1:length(start_indices)
    segment = x(start_indices(k):start_indices(k)+N-1);
    accumulator = accumulator + mperiodogram(segment, N, w);
end
Welch = accumulator / length(start_indices);
```

4.3 Why the estimate is smoother

Each segment contains a different phase and a different local state of the swept and intermittent components. Spectral details that are not repeatable fluctuate from segment to segment, while persistent frequency content remains. The arithmetic mean therefore reduces variance. However, the result merges all temporal states into one spectrum and cannot indicate when a component was active.

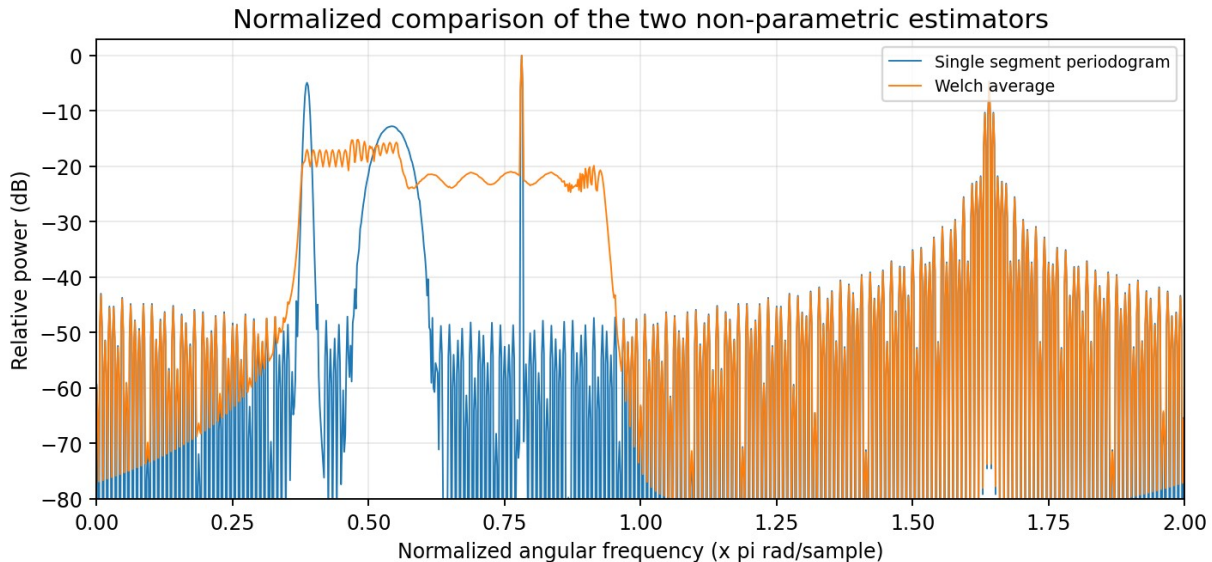


Figure 2. Relative comparison between a single-segment periodogram and the full-record Welch estimate.

5. Short-time spectral analysis

5.1 Parameter conversion

The frequency-grid spacing is supplied as $df = 2\pi/N$. The function reconstructs the FFT length with $N = \text{round}(2\pi/df)$. The time parameter dt is the hop between consecutive frames. For $N = 1024$ and $dt = 128$, the overlap is 896 samples, or 87.5 percent.

5.2 Matrix construction

```
start_indices = 1:dt:(length(x) - N + 1);
raw_periodograms = zeros(N, length(start_indices));
for k = 1:length(start_indices)
    frame = x(start_indices(k):start_indices(k)+N-1);
    raw_periodograms(:,k) = mperiodogram(frame, N, w);
end
```

5.3 Variance parameter

The original interface includes `rvar`. In the completed implementation, `rvar = 1` returns the unsmoothed short-time periodograms. Values above one average each column with the previous `rvar - 1` columns. This is a simple Welch-like temporal average that preserves the N -bin grid while reducing variance, although it also blurs fast changes.

5.4 Output dimensions

The 32768-sample record provides 249 complete frames: $1 + (32768 - 1024)/128 = 249$. Consequently, `P_stft` has 1024 rows and 249 columns. Frame centres are reported in sample indices for an interpretable horizontal axis.

Resolution trade-off. The 1024-sample window resolves closely spaced frequencies, but it spans four complete 256-sample ON/OFF cycles of S2. The intermittent tone therefore appears much more persistent than it would with a shorter window.

6. Synthetic signal model

The complete signal is $x[n] = s1[n] + s2[n] + s3[n] + s4[n]$. All components are complex exponentials, which avoids mirror images caused by real-valued sinusoids and makes each designed digital frequency appear at a single FFT bin.

Component	Type	Frequency / range	Period	Purpose
S1	Constant tone	bin 400	-	Stationary reference
S2	Gated tone	bin 840	256 samples	Intermittent component, 58.6% duty cycle
S3	Asymmetric triangular sweep	bins 240-480	4096 samples	Bidirectional non-stationary track
S4	Rising sawtooth sweep	bins 192-288	8192 samples	Repeated upward chirp

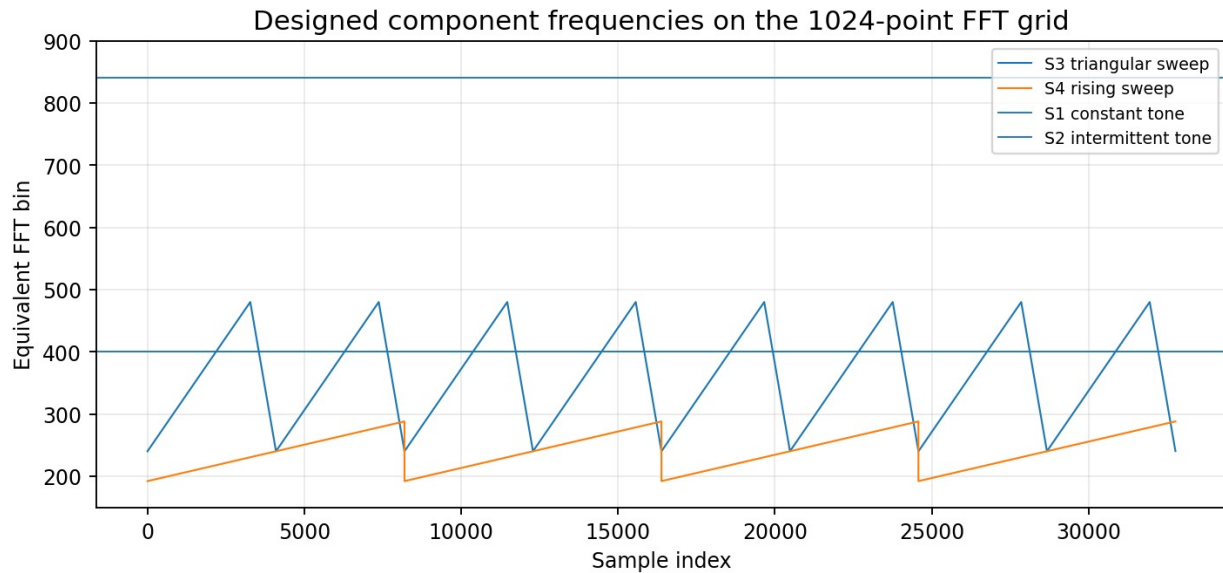


Figure 3. Theoretical component frequencies expressed on the 1024-point FFT grid.

6.1 Phase construction

For the swept components, the instantaneous angular frequency is integrated with a cumulative sum to obtain phase. The resulting sample is $\exp(j \text{ times cumulative phase})$. This method directly controls the local frequency trajectory and is suitable for testing time-frequency algorithms.

7. Main processing workflow

7.1 Analysis configuration

```
N = 1024;
w = 0.5 - 0.5*cos(2*pi*(0:N-1)/(N-1));
P_periodogram = mperiodogram(x, N, w);
P_welch = mwelch(x, N, 512, w);
dt = 128;
df = 2*pi/N;
P_stft = mstft(x, dt, df, 1, w);
```

7.2 Axis definitions

The one-dimensional plots use normalized angular frequency $2k/N$, expressed as multiples of π rad/sample. The spectrogram uses FFT-bin number vertically and frame-centre sample index horizontally. This avoids ambiguity between digital frequency, physical frequency and array index.

7.3 Decibel conversion

All power values are converted using $10 \log_{10}(P)$. The $\max(P, \text{eps})$ safeguard prevents $\log_{10}(0)$ from producing negative infinity. This does not alter meaningful bins because eps is far below the observed numerical power levels.

7.4 Reproducibility

The script saves the signal components, spectral estimates and axes in `practice_results.mat`. The ZIP additionally contains `reference_results.mat`, generated by an independent NumPy implementation with the same equations. This allows array dimensions and dominant-bin locations to be checked before or after running MATLAB.

Toolbox independence. The supplied code constructs the Hann window, asymmetric triangle and rising sawtooth explicitly and performs segmentation with index arithmetic. The core practice therefore does not depend on `buffer`, `hann`, `sawtooth`, `periodogram`, `pwelch` or `spectrogram`.

8. Periodogram results

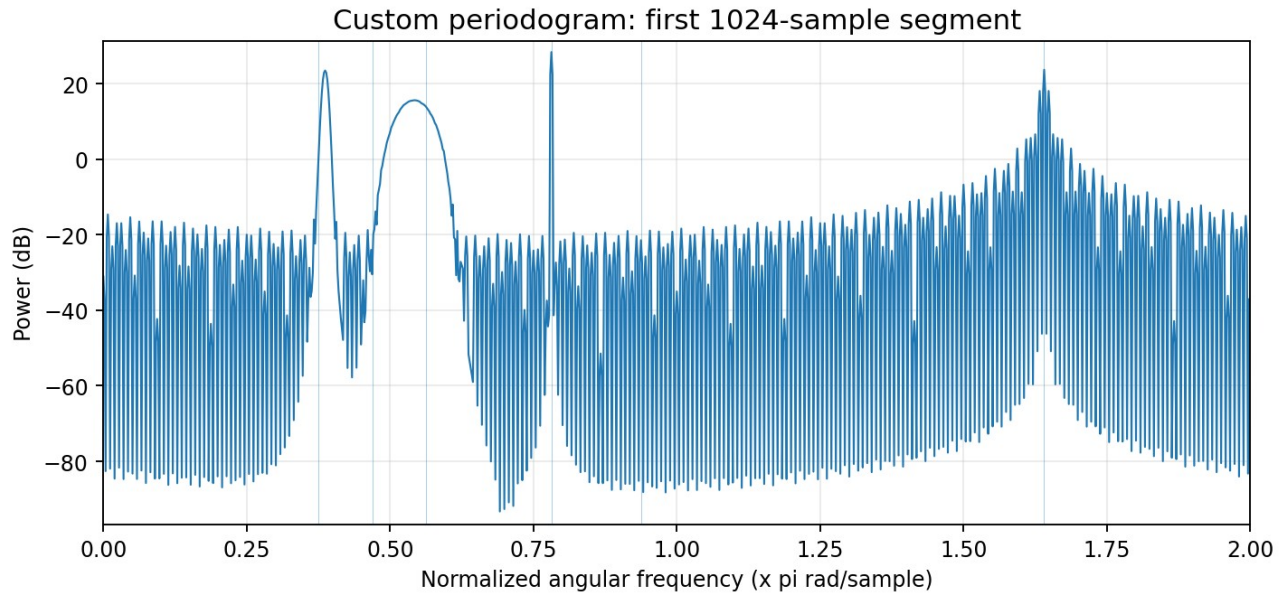


Figure 4. Periodogram obtained from only the first 1024 samples.

8.1 Observations

- The constant tone produces the dominant narrow peak at FFT bin 400.
- The intermittent tone contributes energy around bin 840, but time gating creates sidebands and leakage.
- Only the initial portions of S3 and S4 are represented, so the periodogram cannot describe their complete sweep ranges.
- The Hann window suppresses distant leakage but broadens every local spectral peak.

8.2 Main limitation

A single periodogram is a snapshot of one selected segment. It is not a global characterization of the record, and it has no mechanism to associate spectral content with time. This is why its title explicitly states that only the beginning of the record is shown.

9. Welch results

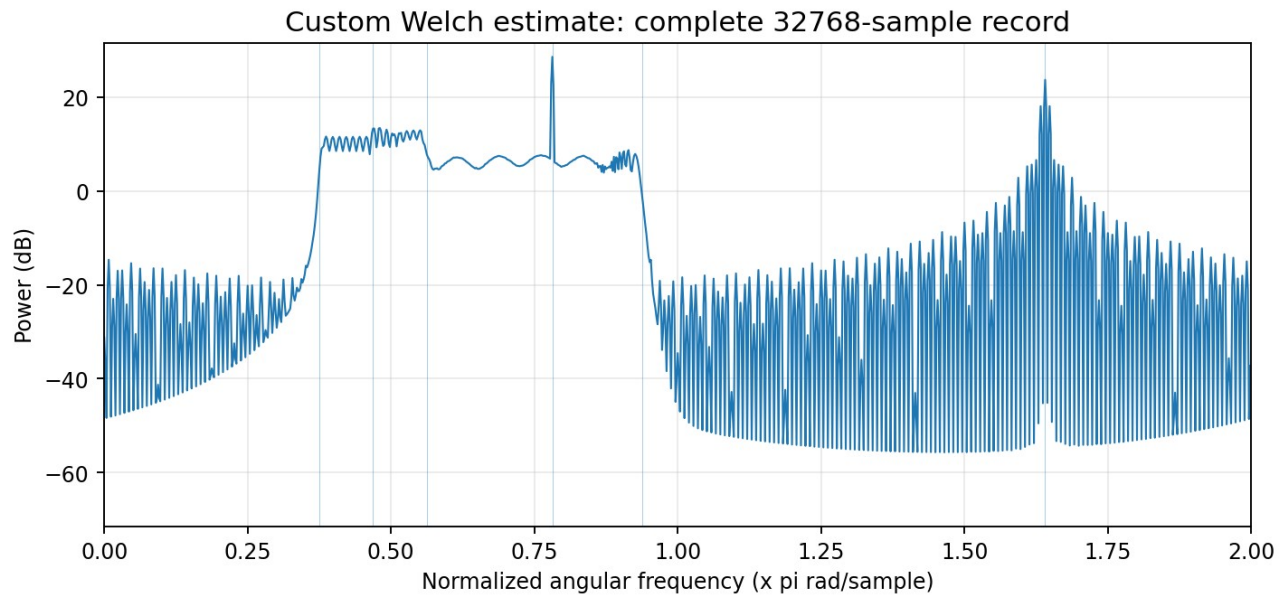


Figure 5. Welch average over the complete signal record.

9.1 Observations

- The fixed line at bin 400 remains sharp and dominant because it is present in every segment.
- The gated tone is visible around bin 840 with lower average power because it is active for only 150 of every 256 samples.
- S3 distributes energy throughout approximately bins 240 to 480.
- S4 fills approximately bins 192 to 288.
- The curve is substantially more stable than a single periodogram because 63 estimates are averaged.

9.2 Lost information

Although Welch reveals the frequency occupancy of all four components, it collapses the complete record along the time dimension. A triangular sweep, a random hopping tone and a collection of stationary tones could produce similar broad global spectra. Their different dynamics can only be distinguished with a time-frequency representation.

10. STFT results

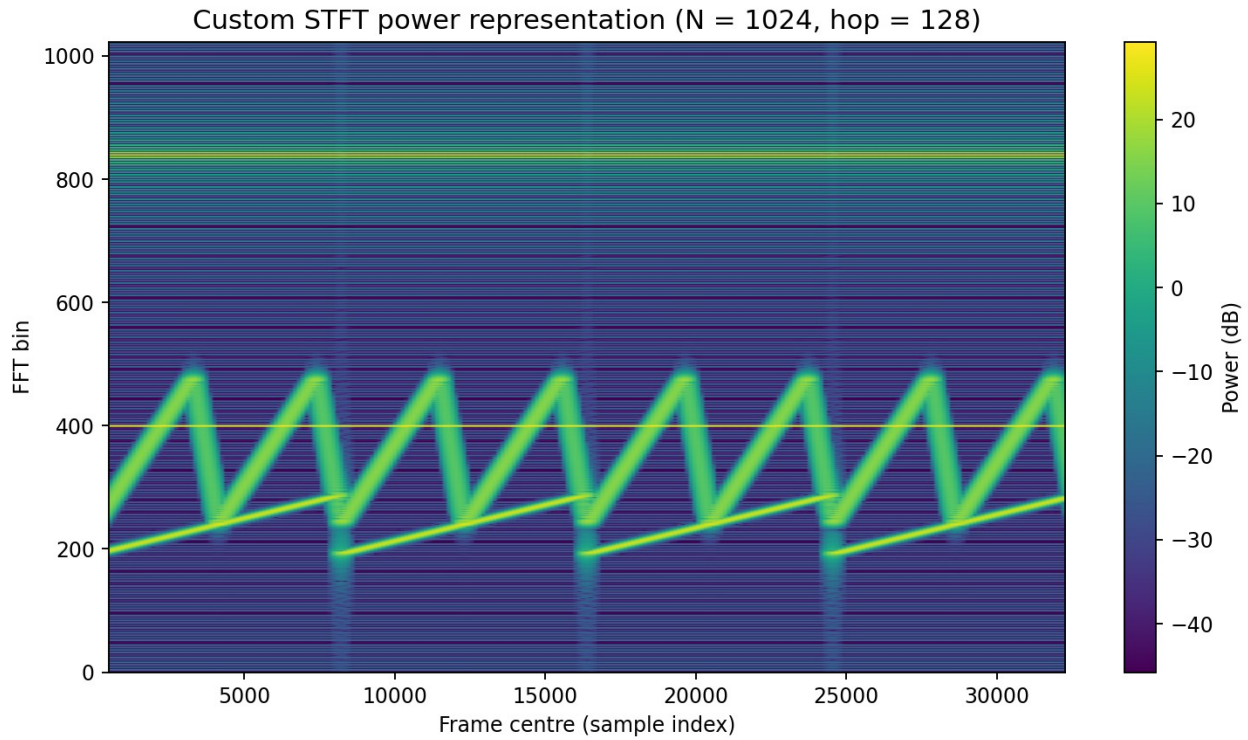


Figure 6. Full custom short-time power representation.

10.1 Interpretation

The horizontal line at bin 400 corresponds to S1. The high-frequency horizontal structure near bin 840 corresponds to S2. The repeated diagonal and triangular trajectories in the lower half are generated by S3 and S4. Unlike Welch, the STFT retains the order in which these frequencies occur.

10.2 Time-frequency compromise

With a 1024-sample window, the chirp tracks have good frequency definition but are averaged over a relatively long interval. The 128-sample hop densely samples the time axis, yet overlapping frames do not create new intrinsic time resolution; they only provide smoother tracking of the windowed estimate.

11. Three-dimensional representation

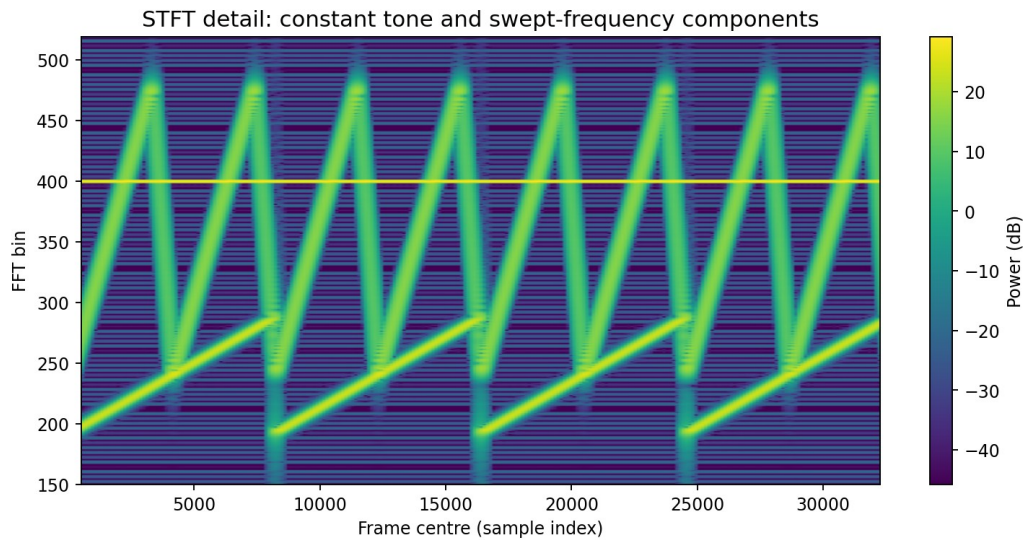


Figure 7. Expanded STFT region containing S1, S3 and S4.

The detail view separates the stationary bin-400 ridge from the two swept components. Crossings do not imply physical interaction; they only indicate that multiple components momentarily occupy similar frequency bins.

Three-dimensional time-frequency power surface

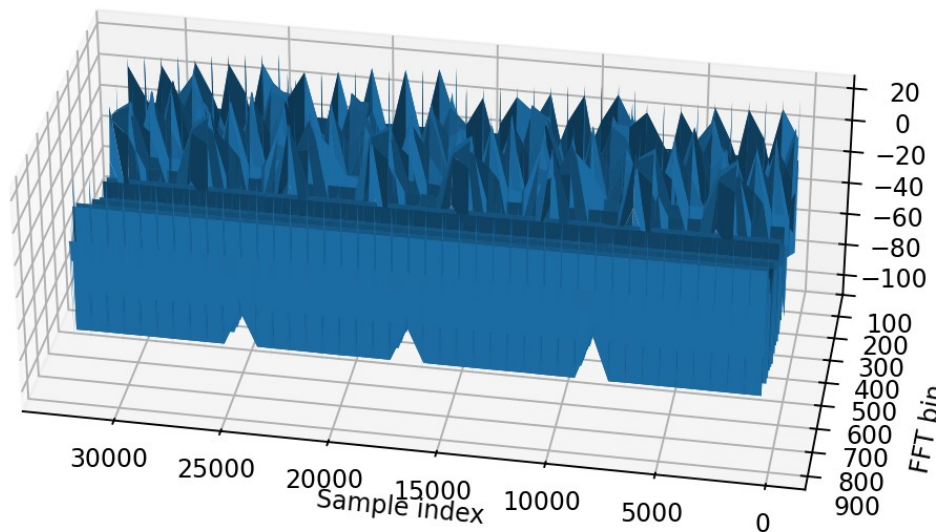


Figure 8. Decimated three-dimensional surface of the short-time power matrix.

The surface emphasizes relative ridge power, while the two-dimensional image remains more precise for reading coordinates. Plotting is decimated, but the stored matrix retains all 1024 by 249 values.

12. Discussion

12.1 Comparison of estimators

Criterion	Periodogram	Welch	STFT
Time localization	One selected segment	None	One spectrum per frame
Variance	High	Lower through averaging	High per frame; optional smoothing
Frequency resolution	Set by N and window	Set by N and window	Set by N and window
Best use here	Inspect the beginning	Global frequency occupancy	Track dynamic components

12.2 Intermittent-tone visibility

S2 switches every 256 samples, while each STFT frame spans 1024 samples. Every complete frame therefore contains roughly four switching cycles. The line near bin 840 is consequently present in most frames with an averaged amplitude. Reducing N would expose the ON/OFF pattern more clearly but would widen frequency peaks.

12.3 Overlap interpretation

Welch uses 50 percent overlap to increase the number of averages without excessive redundancy. The STFT uses 87.5 percent overlap because its purpose is to sample smooth time trajectories. Overlap does not independently improve the fundamental resolution imposed by the window duration.

12.4 Complex-signal convention

The frequency axis is not fftshifted. Frequencies are displayed from zero to almost 2π , so the negative-frequency tone at $-23/128$ cycles/sample appears at the equivalent positive-grid location $105/128$, namely bin 840.

13. Numerical validation and complexity

13.1 Independent numerical reproduction

The packaged reference data were generated with an independent NumPy implementation of the same equations. This is not a substitute for executing MATLAB, but it verifies the signal construction, array dimensions and dominant frequency locations without using MATLAB spectral-estimation functions.

Check	Result
Signal length	32768 samples
Periodogram size	1024 x 1
Welch segments	63
Welch output size	1024 x 1
STFT frames	249
STFT output size	1024 x 249
Strongest periodogram bin	400
Strongest Welch bin	400

13.2 Computational cost

A single N -point periodogram requires one FFT, with approximate cost $O(N \log N)$. Welch requires K FFTs, giving $O(K N \log N)$, where $K = 63$. The STFT requires M FFTs, giving $O(M N \log N)$, where $M = 249$. Its main stored array contains 254,976 power values. Increasing overlap increases M linearly and therefore increases both computation and storage.

13.3 Numerical stability

The algorithms use double-precision arithmetic by default. Window normalization avoids arbitrary gain changes between windows, and the decibel conversion is protected with `eps`. Because the test signal is deterministic and contains no additive noise, the visible background mainly represents finite-window leakage and numerical round-off.

14. Limitations and improvements

- The implemented scaling is a discrete power spectrum. To report a physical power spectral density, divide by the sampling frequency and state the units explicitly.
- The STFT function uses complete frames only. Zero-padding the final partial frame could extend the plot to the exact end of a record, but the added samples would need to be documented.
- Temporal averaging through `rvar` is a practical extension of the supplied interface, not a unique universal definition of variance reduction.
- The current visualization is unshifted. An `fftshift` option would make negative frequencies more intuitive by displaying the interval from $-\pi$ to π .
- Peak interpolation could estimate frequencies between FFT bins, while reassignment or synchrosqueezing could sharpen chirp ridges.
- Adding controlled noise and repeated Monte Carlo trials would permit a quantitative bias-variance comparison.

14.1 Recommended alternative configuration

To resolve the 150-sample ON intervals of S2, a shorter window such as $N = 128$ or $N = 256$ is appropriate. The corresponding frequency peaks would be wider, illustrating the uncertainty trade-off directly. A useful extension is to produce two spectrograms with different N values and compare their ability to separate frequency and time events.

14.2 Verification in MATLAB

After running the supplied main script, the custom outputs can be compared with MATLAB reference functions such as `periodogram`, `pwelch` and `spectrogram` using identical windows, overlap, FFT lengths and two-sided settings. Small differences may result from endpoint conventions, detrending options and density normalization, so those parameters must be matched before comparing arrays.

Important. This report does not claim that MATLAB itself was executed in the present environment. The numerical figures and MAT reference file were produced independently from the documented equations.

15. Conclusions and references

15.1 Conclusions

The practice successfully implements three complementary non-parametric spectral tools. The periodogram gives a high-resolution but high-variance view of one segment. Welch averaging provides a stable global spectrum and reveals the frequency occupancy of the complete record, but removes temporal ordering. The short-time representation preserves that ordering and clearly identifies stationary ridges and swept-frequency trajectories. The experiment also shows that overlap controls the sampling of adjacent frames, while window duration determines the fundamental time-frequency compromise.

The final MATLAB code is self-contained at the algorithmic level: it constructs the test waveforms and Hann window explicitly and does not call prebuilt periodogram, Welch or spectrogram estimators. Input checks, explicit axes, protected logarithms and MAT-file export improve robustness and reproducibility.

15.2 References

- [1] P. D. Welch, "The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms," IEEE Transactions on Audio and Electroacoustics, vol. 15, no. 2, pp. 70-73, 1967.
- [2] F. J. Harris, "On the use of windows for harmonic analysis with the discrete Fourier transform," Proceedings of the IEEE, vol. 66, no. 1, pp. 51-83, 1978.
- [3] A. V. Oppenheim and R. W. Schaffer, Discrete-Time Signal Processing, 3rd ed., Pearson, 2010.
- [4] L. Cohen, Time-Frequency Analysis, Prentice Hall, 1995.

15.3 Archive contents

The archive root contains README.txt. The code folder contains the four .m files. The report folder contains this PDF. The data folder contains reference_results.mat and validation_summary.txt. The figures folder contains the images used in the report.